

METU Computer Engineering

CEng 111 – Fall 2026
Week 2b

Credit: Some slides are from the “Invitation to Computer Science” book by G. M. Schneider, J. L. Gersting and some from the “Digital Design” book by M. M. Mano and M. D. Ciletti.

A computer

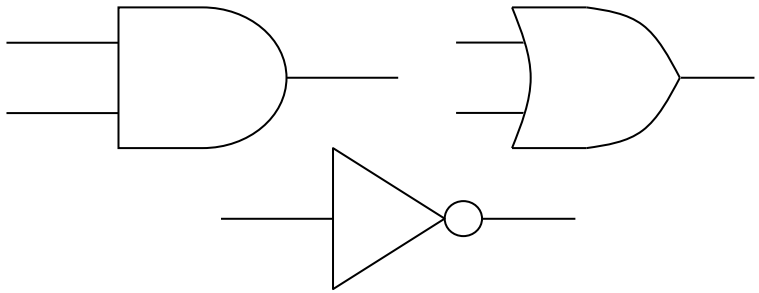


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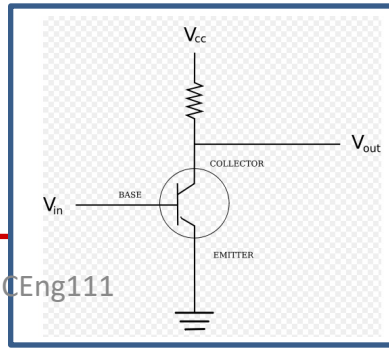




Devices



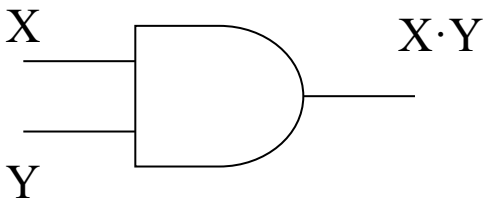
Gates



Transistors

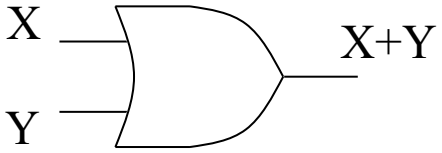
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AND gate



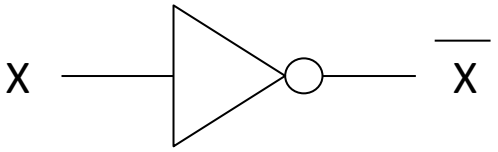
X	Y	X·Y
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate



X	Y	X+Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT Gate



X	$\overline{X}$
0	1
1	0

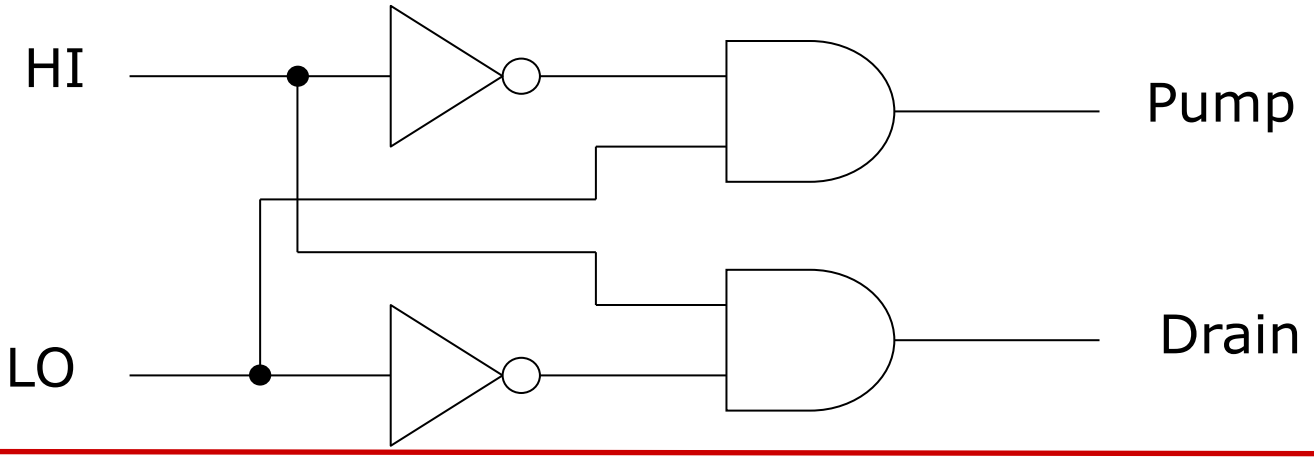
An example problem: Water Tank

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Truth Table Representation

HI	LO	Pump	Drain	
0	0	0	0	→ Tank level is OK
0	1	**1**	0	→ Low level, pump more in
1	0	0	**1**	→ High level, drain some out
1	1	x	x	→ Inputs cannot occur

Schematic Representation

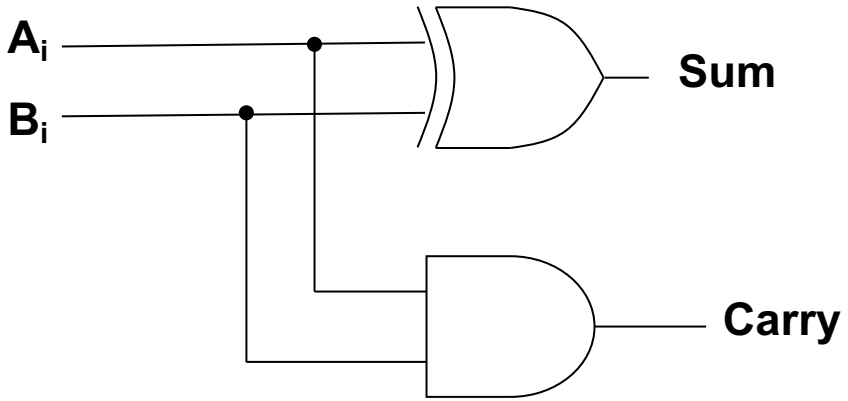


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1-bit Half-adder

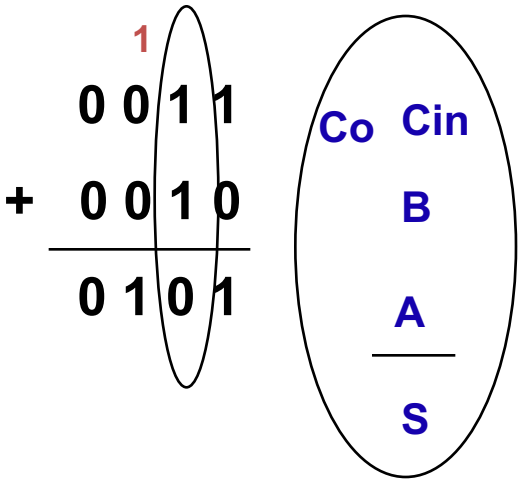
$$\begin{array}{rrrr} 0 & 1 & 0 & 1 \\ +\,0 & +\,0 & +\,1 & +\,1 \\ \hline 0 & 1 & 1 & 10 \end{array}$$

A_i	B_i	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



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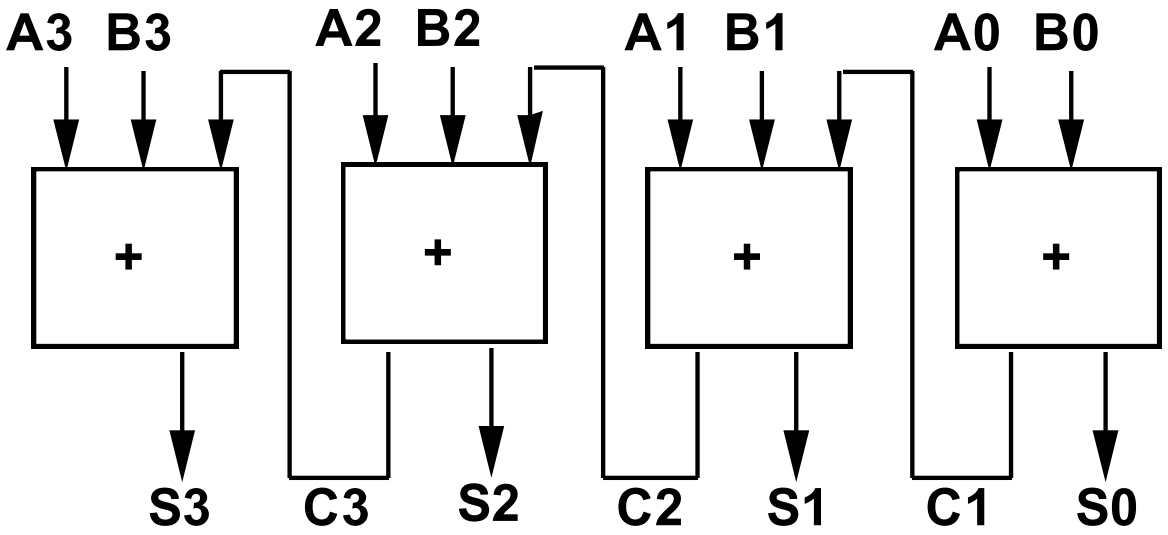
1-bit full-adder



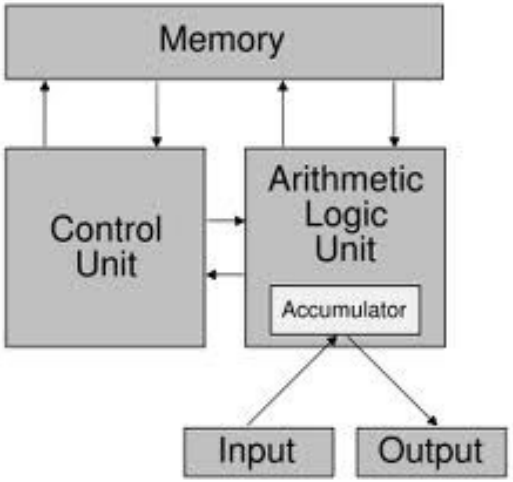
A	B	CI	S	CO
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

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N-bit Adder



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Computer Organization

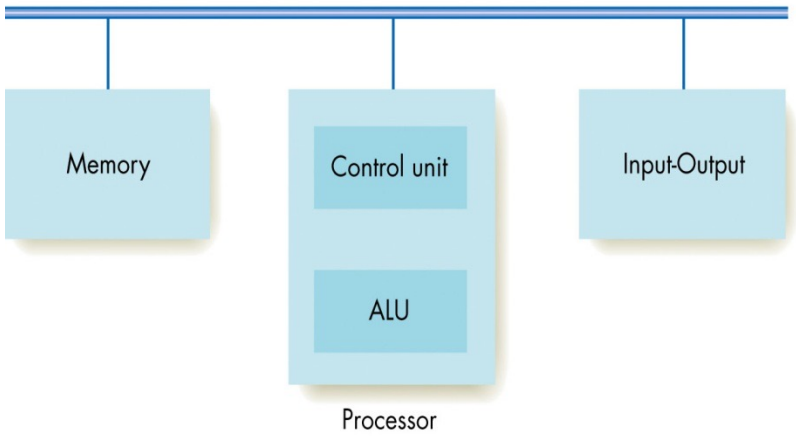
VON NEUMANN ARCHITECTURE & ITS IMPLEMENTATION

The Components of a Computer System

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- Von Neumann architecture has four functional units:
  - Memory
  - Input/Output
  - Arithmetic/Logic unit
  - Control unit



- Sequential execution of instructions
- Stored program concept



Today

- Digital Computation:
  - Von Neumann Architecture
  - Memory
  - CPU
    - Control Unit and the Fetch-Decode-Execute Cycle



Administrative Notes

- History of computing quiz
  - Material shared
  - Online quiz:
    - Monday (12 Oct) between 19:00 and 19:30
    - On ODTUClass page of the course.
- Submitting busy hours
  - Deadline: 9 Oct 23:55
- Linux Workshops:
  - 7 Oct 18:00 & 13 Oct 18:30

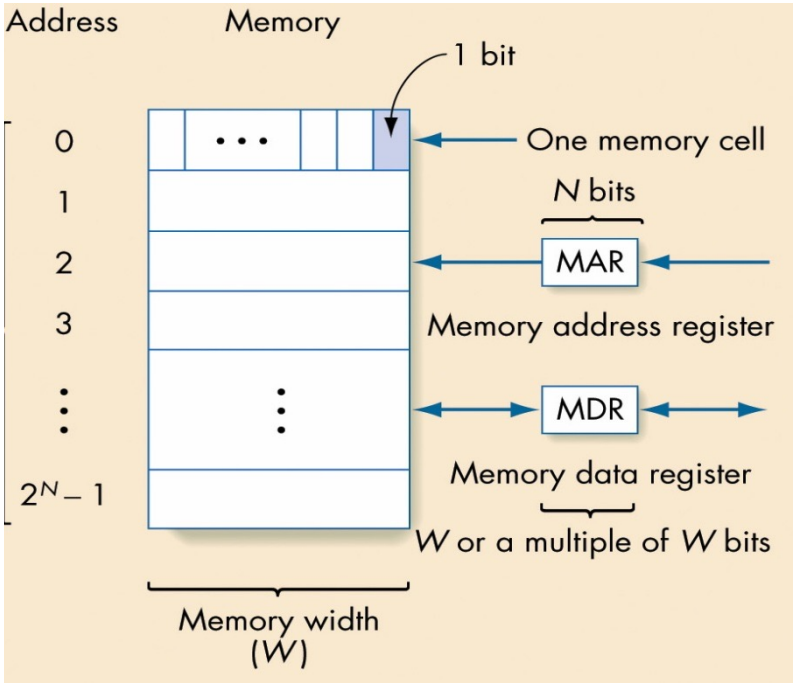


Memory and Cache

- Information is stored and fetched from memory subsystem
- Memory maps addresses to memory locations
- Cache memory keeps values currently in use in faster memory to speed access times

Memory and Cache (continued)

- RAM (Random Access Memory)
  Often called *memory*, *primary memory*
  - Memory made of addressable "cells"
  - Cell size is 8 bits
    - Nowadays, it is 32 or 64 bits.
  - All memory cells accessed in equal time
  - Memory address
    - Unsigned binary number with N bits
    - Address space is then 2^N cells



Memory Unit	Description
Bit	Binary Digit 1 or 0
Byte	8 bits
KiloByte (KB)	1024 Bytes
MegaByte (MB)	1024 KB
GigaByte (GB)	1024 MB
TeraByte (TB)	1024 GB
PetaByte (PB)	1024 TB
HexaByte or exaByte (EB)	1024 PB
ZettaByte (ZB)	1024 EB
YottaByte (YB)	1024 ZB
BrontoByte	1024 YB
GeopByte	1024 Bronto Bytes

Table: https://www.krivalar.com/computer-memory-units



Memory and Cache (continued)

- Rapid access, low capacity “warehouse”
- Retains information entered through input unit
- Retains info that has already been processed until can be sent to output unit

Memory and Cache (continued)

- Parts of the memory subsystem
  - Fetch/store (or Read/Write) controller
    - Fetch: retrieve a value from memory
    - Store: store a value into memory
  - Memory address register (MAR)
  - Memory data register (MDR)



Memory and Cache (continued)

- Fetch operation
  - The address of the desired memory cell is moved into the MAR
  - Fetch/store controller signals a “fetch,” accessing the memory cell
  - The value at the MAR’s location flows into the MDR



Memory and Cache (continued)

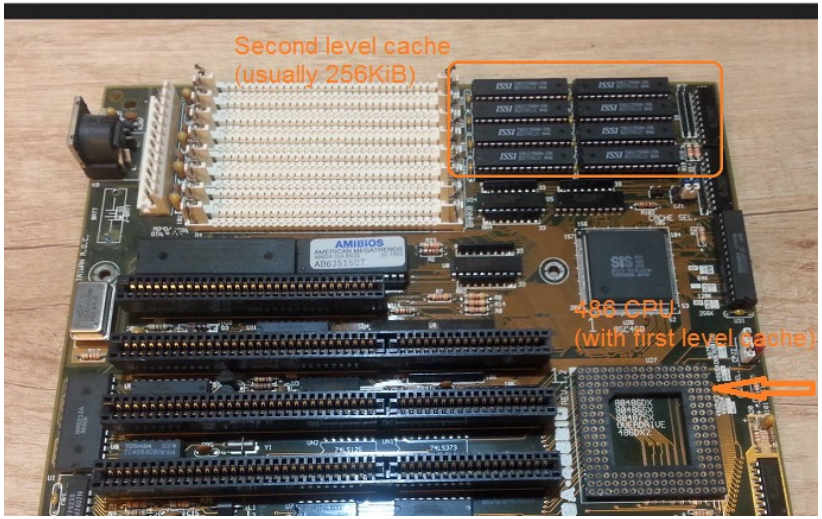
- Store operation
  - The address of the cell where the value should go is placed in the MAR
  - The new value is placed in the MDR
  - Fetch/store controller signals a “store,” copying the MDR’s value into the desired cell



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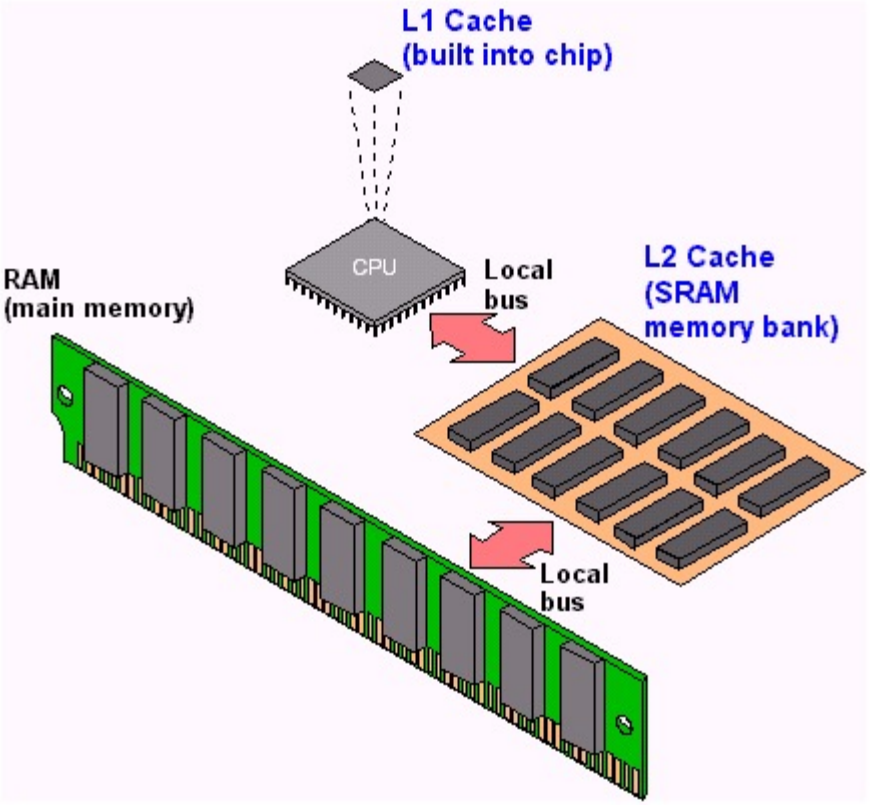
Cache Memory

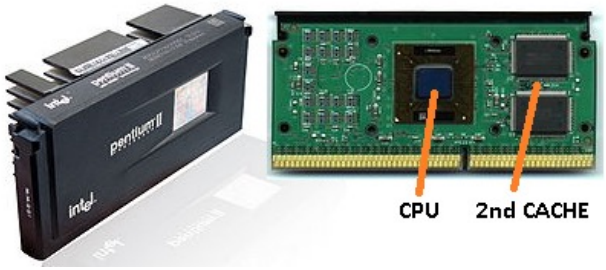
- Memory access is much slower than processing time
- Faster memory is too expensive to use for all memory cells
- Locality principle
  - Once a value is used, it is likely to be used again
- Small size, fast memory just for values currently in use speeds computing time

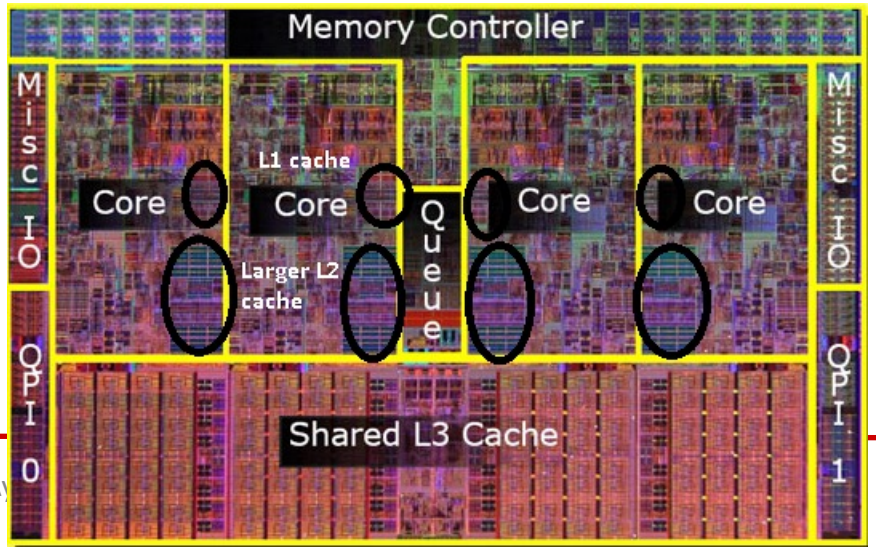


80486: (1989)

From Computer Desktop Encyclopedia
© 1999 The Computer Language Co. Inc.

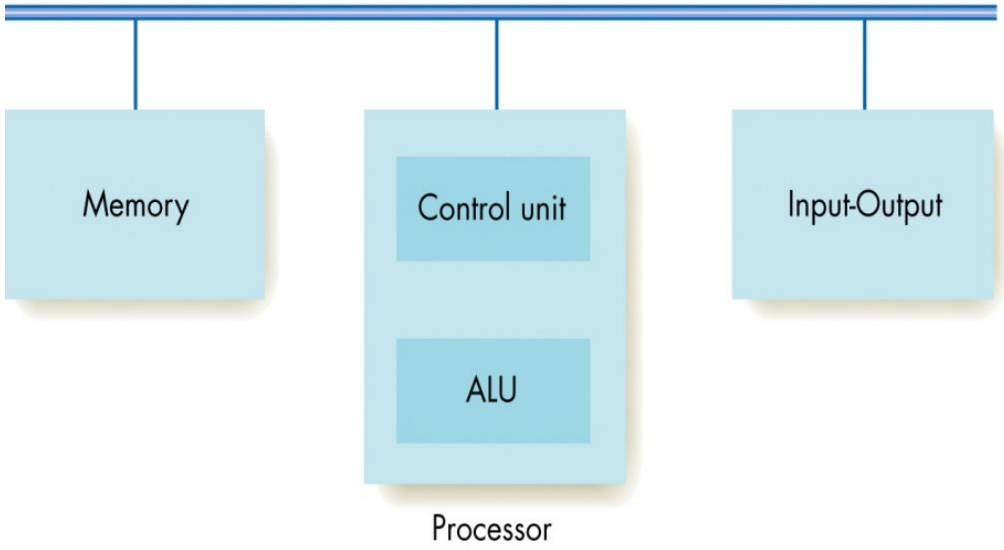






Central Processing Unit (CPU)

- Control Unit
- Arithmetic Logic Unit (ALU)



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The Control Unit



- Manages stored program execution
- Task

1. Fetch from memory the next instruction to be executed
2. Decode it: determine what is to be done
3. Execute it: issue appropriate command to ALU, memory, and I/O controllers



Machine Language Instructions

- Can be decoded and executed by control unit
- Parts of instructions
  - Operation code (op code)
    - Unique unsigned-integer code assigned to each machine language operation
  - Address field(s)
    - Memory addresses of the values on which operation will work



Typical Machine Language Instruction Format

More on Instructions



- LOAD X -> Load register R with the contents of memory cell X
- STORE X -> Store register R into memory cell X
- MOVE X, Y -> Copy the contents of X into Y
- ADD X -> Add contents of X to the contents of R
- ADD X,Y -> Add contents of X to the contents of Y, and put the result in register R
- COMPARE X, Y-> Set GT (greater than), EQ (equal) and LT (less than) condition codes
- JUMP X-> Jump unconditionally to the instruction in cell X
- JUMPGT X-> Jump, if GT=1, to the instruction in cell X
- HALT



Machine Language Instructions (continued)

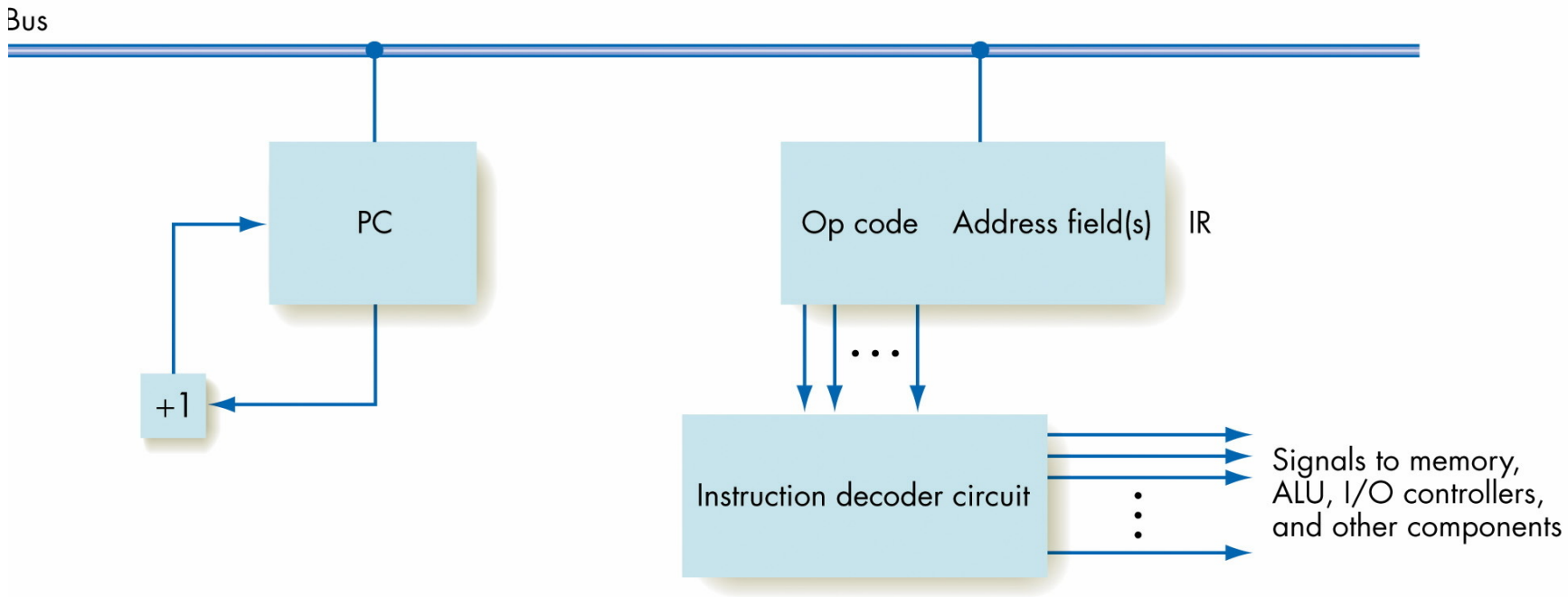
- Types of machine instructions:
  - Data transfer
    - Move values to and from memory and registers
  - Arithmetic/logic
    - Perform ALU operations that produce numeric values
  - Compares
    - Set bits of compare register to hold result
  - Branches
    - Jump to a new memory address to continue processing



Control Unit Registers and Circuits

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- Parts of control unit
  - Linked to other subsystems
  - Instruction decoder circuit
  - Two special registers:
    - Program Counter (PC)
      - Stores the memory address of the next instruction to be executed
    - Instruction Register (IR)
      - Stores the code for the current instruction



Organization of the Control Unit Registers and Circuits